

NrgOpt Photovoltaic Project Site Survey Checklist

Project Information

Item	Detail
Project / Owner Name	
Project Address	
GPS Coordinates (lat/lng)	
Owner Contact Person	
Owner Contact Phone	
Survey Date	
Survey Engineer	
Weather During Survey	

1. Building Specifications

1.1 Basic Dimensions

- ☐ Building area (m²): _
- ☐ Elevation (m): _
- ☐ Number of floors: _
- ☐ Building orientation: _
- ☐ Year built / time in use: _
- ☐ Site usage / ownership type: _

1.2 Structural Load

- ☐ Roof design live load (kg/m²): _
- ☐ Current load status: _
- ☐ Remaining capacity (kg/m²): _
- ☐ Structural reinforcement required? ☐ Yes ☐ No
- ☐ Building seismic rating: _
- ☐ Structural report available? ☐ Yes ☐ No

1.3 Roof Type

- ☐ **Concrete Roof:** LxWxH(m), parapet height(m), openings, pitch, leakage, repairs, skylights, obstacles(LxWxH), waterproofing
- ☐ **Metal Sheet Roof:** Profile (standing seam/angle/trapezoidal), pitch, beam/purlin/column specs & spans, rust condition, panel thickness(mm), drawing accuracy
- ☐ Roof access: ☐ Stairs ☐ Ladder ☐ Hatch

1.4 Obstruction Survey

- ☐ Obstruction 1: L__ x W__ x H (m), Direction: ____
- ☐ Obstruction 2: L__ x W__ x H (m), Direction: ____
- ☐ Obstruction 3: L__ x W__ x H (m), Direction: ____

1.5 Shading Analysis

- ☐ Surrounding building 1: H__m, D__m, Direction ____
- ☐ Surrounding building 2: H__m, D__m, Direction ____
- ☐ Trees or vegetation shading? ☐ Yes ☐ No
- ☐ Nearby chimneys/exhaust pipes? ☐ Yes ☐ No (H: __m, D: __m)
- ☐ Winter solstice sun path checked? ☐ Yes ☐ No
- ☐ Drone shadow simulation completed? ☐ Yes ☐ No

1.6 Solar Panel Layout

- ☐ Rail spacing (mm): _ Tilt angle (°): _ Row spacing (m): _
- ☐ Orientation: ☐ Landscape ☐ Portrait Est. panel count: _
- ☐ PV station height (m): _

2. Location & Solar Data

Parameter	Value	Method
Coordinates		☐ Address lookup ☐ GPS
Annual Irradiance (kWh/m ²)		
Equivalent Hours (h)		
Optimal Tilt Angle (°)		
Suggested Efficiency (%)		

3. Meteorological & Environmental

3.1 Climate Data

- ☐ Annual irradiation (kWh/m²): _ Max wind speed (m/s): _
- ☐ Snow load (kN/m²): _ Extreme low (°C): _ Extreme high (°C): _
- ☐ Annual avg temp (°C): _ Annual rainfall (mm): _
- ☐ Lightning frequency: ☐ Low ☐ Medium ☐ High

3.2 Environmental Risks

- ☐ Within 5km of coastline? ☐ Yes ☐ No (salt spray)
- ☐ Nearby chemical/cement/steel plants? ☐ Yes ☐ No (dust)
- ☐ Nearby water bodies (flood risk)? ☐ Yes ☐ No
- ☐ Earthquake-prone zone? ☐ Yes ☐ No
- ☐ Typhoon path? ☐ Yes ☐ No
- ☐ Soil corrosion: ☐ Low ☐ Medium ☐ High
- ☐ Bird activity: ☐ Low ☐ Medium ☐ High

4. Electrical Infrastructure

4.1 Grid Connection

- ☐ Connection type: ☐ LV ☐ HV
- ☐ Transformer count: _ Capacity (kVA): _ Model: _
- ☐ Protection devices (model): _ Existing PV capacity (kW): _
- ☐ Distance to connection point (m): _

4.2 Distribution Room

- ☐ Location / photo taken: ☐ Yes ☐ No
- ☐ Space for PV cabinet? ☐ Yes ☐ No Spare breakers: _
- ☐ Cable trench? ☐ Yes ☐ No Road crossing required? ☐ Yes ☐ No
- ☐ Single-line diagram available? ☐ Yes ☐ No

4.3 Consumption Profile

- ☐ Generation mode: ☐ Full export ☐ Self-consumption + surplus
- ☐ Contracted demand (kW): _ Peak period: _:00 - _:00
- ☐ Weekend/holiday operation? ☐ Yes ☐ No Night shift? ☐ Yes ☐ No

4.4 Monthly Consumption (kWh)

Mo	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
kWh												

Annual total: _ kWh

4.5 Tariff Information

- ☐ Supply company: _ Category: ☐ Industrial ☐ Commercial ☐ Other
- ☐ Weighted avg rate (CNY/kWh): _ Demand charge (CNY/kW/mo): _

4.6 Time-of-Use Pricing

Period	Rate (CNY/kWh)	Start	End
Critical Peak			
Peak			
Shoulder			
Off-Peak			

5. Site Access & Logistics

5.1 Access Assessment

- ☐ Road width (m): _ Max truck length (m): _
- ☐ Crane radius (m): _ Parking position confirmed? ☐ Yes ☐ No
- ☐ Road closure/permit required? ☐ Yes ☐ No Staging area? ☐ Yes ☐ No
- ☐ Material lifting: ☐ Elevator ☐ Crane ☐ Other: _
- ☐ Construction power/water available? ☐ Yes ☐ No

5.2 Safety Considerations

- ☐ Roof edge protection? ☐ Yes ☐ No Roof access type: _
- ☐ Fall protection anchors? ☐ Yes ☐ No Evacuation route? ☐ Yes ☐ No
- ☐ Asbestos/hazardous materials? ☐ Yes ☐ No

6. Regulatory & Documentation

6.1 Owner Documentation

- ☐ Property ownership ☐ Land use certificate ☐ Construction drawings ☐ Structural report
- ☐ Electrical single-line ☐ Fire safety ☐ EIA ☐ Business license

6.2 Regulatory Check

- ☐ Local PV filing requirements: _
- ☐ Grid pre-approval needed? ☐ Yes ☐ No Height restriction? ☐ Yes ☐ No
- ☐ Heritage/conservation area? ☐ Yes ☐ No Aviation restriction? ☐ Yes ☐ No

7. Photo Checklist

#	Item	✓	#	Item	✓
1	North view from roof		11	Distribution room interior	
2	East view from roof		12	Spare breaker slots	
3	South view from roof		13	Transformer nameplate	
4	West view from roof		14	Cable route	
5	Roof panorama (drone)		15	Site entrance	
6	Roof close-up x 4		16	Surrounding buildings	
7	Obstructions close-up		17	Roof access point	
8	Parapet/roof edge		18	Drone video	
9	Existing rooftop equipment		19	GPS-tagged photo	
10	Distribution room exterior		20	Team photo	

8. Cost & Timeline

- ☐ Est. cable length (m): _ Est. trenching (m): _
- ☐ Special access equipment? ☐ Yes () ☐ No
- ☐ Grid outage window: _ Construction start: _ Duration (days): _

9. Sign-off

Role	Name	Signature	Date
Survey Engineer			
Project Manager			
Owner Representative			
Electrical Engineer			
Structural Engineer			

10. Additional Notes